Earth Science Informatics (2026) 19:46
https://doi.org/10.1007/s12145-026-02100-2

RESEARCH

Intelligent identification of rock mass discontinuities: an Attention-PointNet++ approach integrated with forward modeling

Jiateng Guo[1] · Juanli Zhang[1] · Ziming Xiong[2] · Zirui Zhang[1] · Luyuan Wang[1] · Binbin Cheng[1] · Tianhong Yang[1]

Received: 27 November 2025 / Accepted: 4 March 2026 / Published online: 21 March 2026
© The Author(s), under exclusive licence to Springer-Verlag GmbH Germany, part of Springer Nature 2026

Abstract

The precise identification of rock mass discontinuities is a crucial part of the stability analysis of rock mass slopes. Aiming at the problems such as the low efficiency of traditional machine learning methods relying on manually annotated data, and the limitations of the existing PointNet++network in the extraction of point cloud neighborhood features, this study proposes an innovative deep learning method named Attention-PointNet++that integrates the forward modeling of discontinuities and attention enhancement. Its specific process includes: (1) Using the point cloud coordinates and normal vectors as input features, adopting a local area random sampling strategy, and constructing a training set (accounting for 80%) and a test set (accounting for 20%) based on the forward modeling of discontinuities; (2) Introducing the Attention-PointNet++network architecture improved by integrating the self-attention mechanism, and strengthening the ability to capture complex geometric features through multi-scale feature fusion; (3) Based on the trained deep learning model, realizing the intelligent identification of rock mass discontinuities, and completing the extraction of a single discontinuity and the calculation of orientation parameters by combining the density-based spatial clustering of applications with noise (DBSCAN) algorithm and random sample consensus (RANSAC) algorithms. Experiments show that in the standard dataset, the identification results of the three groups of discontinuities are basically consistent with the known data, and for the measured point clouds of two typical engineering cases, this method also shows good robustness. This study provides a new technical solution for the automatic identification of discontinuities in complex geological environments.

Keywords Forward modeling · PointNet++ · Self-attention · Identification of rock mass discontinuities · 3D point cloud

Introduction

As a typical geological body in a complex geological environment, rock mass is a discontinuous medium system jointly composed of various types of rocks and the internal weak discontinuities within it. From the perspective of rock mass mechanics research, it can be decomposed into two basic elements: structures and discontinuities. The former refers to the rock block units formed by being cut by discontinuities, while the latter includes geological interfaces of different genetic types such as faults, joints, bedding planes, and contacts (Hudson and Priest 1979). These discontinuities play a decisive role in the overall physical and mechanical properties of the rock mass and

Communicated by Hassan Babaie

✉ Jiateng Guo
guojiateng@mail.neu.edu.cn

✉ Ziming Xiong
xzm992311@163.com

Juanli Zhang
zhangjuanli_uli@163.com

Zirui Zhang
zhangzirui1992@163.com

Luyuan Wang
2310420@stu.neu.edu.cn

Binbin Cheng
chengbinbin990816@163.com

Tianhong Yang
yangtianhong@mail.neu.edu.cn

[1] College of Resources and Civil Engineering, Northeastern University, Shenyang 110819, China

[2] State Key Laboratory of Explosion & Impact and Disaster Prevention & Mitigation, Army Engineering University of PLA, Nanjing 210007, China

the stability of rock slopes (Alejano et al. 2011; Zhang et al. 2019).

The traditional methods for obtaining the information of discontinuities in rock mass slopes mainly rely on contact measurement technologies. (Priest and Hudson 1976). Limited by the operation efficiency and labor intensity, traditional methods are difficult to meet the needs of modern engineering for refined data. Especially in large-scale slope engineering projects, the measurement cycle often forms a sharp contradiction with the requirements of the project progress (Gigli and Casagli 2011; Guo et al. 2017).

To address the aforementioned limitations of traditional contact measurement technologies, the modern surveying and mapping technology system has undergone continuous innovation and evolution, and non-contact measurement technologies have emerged accordingly and developed a diversified development pattern, serving as a new technical solution for acquiring spatial information of discontinuities in rock mass slopes (Gomes et al. 2016; Gigli et al. 2014; Zhou et al. 2022; Puniach et al. 2023). Among them, close-range photogrammetry, oblique photogrammetry by unmanned aerial vehicles (UAVs), and 3D laser scanning technology constitute the core technology group for current spatial information collection (Assali et al. 2014; Chen et al. 2016; Mineo et al. 2022). It is worth noting that, relying on its capabilities of obtaining high-density point clouds, high-efficiency in rapid data collection, and advantages of large-scale spatial coverage, the 3D laser scanning technology has demonstrated significant technical advantages in the field of geological disaster monitoring (Wehr and Lohr 1999; Wang et al. 2014; Feng et al. 2021).

Currently, the identification of rock mass discontinuities based on 3D point cloud data has become a research hotspot in the field of engineering geology (Cai et al. 2023). Scholars at home and abroad have carried out systematic research on the feature extraction of point cloud data and the geometric modeling of discontinuities, and formed a relatively complete algorithm system. From the methodological perspective, the existing algorithms can be summarized into three categories: (1) Clustering analysis based on point cloud features: Typical algorithms include fuzzy K-means clustering (Slob et al. 2005; Zhang et al. 2021; Olariu et al. 2008; Vöge et al. 2013; Zhou et al. 2024), fuzzy C-means clustering (Hammah and Curran 1998; Lu et al. 2024), RANSAC (Ferrero et al. 2009; Raguram et al. 2012), and iterative self-organizing data analysis (Zhang et al. 2018; Ding et al. 2018). (2) Region-growing algorithms based on spatial division: mainly including the region-growing method (Wang et al. 2017; Ge et al. 2018; Yi et al. 2023), voxelization technology (Hu et al. 2020; Sun et al. 2021), and the box analysis method (Gigli and Casagli 2011). (3) Intelligent algorithms with multi-feature fusion: Representative algorithms include principal component analysis (PCA) (García-Cortés et al. 2012), kernel density estimation (KDE) (Riquelme et al. 2014; Cardia et al. 2023), Hough transform (Leng et al. 2016; Dube and Zell 2011). Current mainstream algorithms still rely on manually set prior rules, such as normal vector thresholds and plane error tolerances, and their intelligence level is difficult to meet the needs of complex engineering scenarios.

In recent years, remarkable breakthroughs in the field of artificial intelligence have brought innovative solutions to the challenging problem of identifying rock mass discontinuities. In related research, machine learning algorithms have begun to emerge and are widely applied to the identification of rock mass discontinuities (Zhou et al. 2025). For example, Ge et al. (2022) applied the ANN algorithm. Through carefully and manually selecting a sufficient training dataset for targeted training, he achieved accurate classification of point cloud data. By leveraging the PointNet and DBSCAN algorithms, Battulwar et al. (2020) successfully achieved the automatic identification of rock mass discontinuities. Ge et al. (2025) also used the GoogLeNet network to build a CNN model, and achieved certain results in improving the identification efficiency. However, to train a network model with excellent performance, it is usually necessary to repeatedly and manually select samples and continuously adjust them until the accuracy requirements are met. This process is extremely cumbersome, seriously restricting the improvement of the automation level. The computational efficiency is also greatly reduced, making it difficult to meet the needs of large-scale and high-efficiency engineering applications.

Aiming at the problems such as the low efficiency of traditional machine learning methods relying on manually annotated data and the limitations of the existing PointNet++network in the extraction of point cloud neighborhood features, this study proposes an innovative deep learning method called Attention-PointNet++, which integrates the forward modeling of discontinuities and attention enhancement. By randomly sampling local regions, a forward modeling of discontinuities is constructed to automatically generate a point cloud dataset of discontinuities with accurate labels. Based on the PointNet++framework, a self-attention mechanism is incorporated to build the Attention-PointNet++network, which is then used to identify real discontinuities. A standard dataset and two point cloud datasets of discontinuities from real-world scenarios are selected as case studies to verify the reliability and effectiveness of the proposed method.

Methodology

The method proposed in this study includes six steps: data collection, data processing, datasets generation, deep learning model building, parameters calculation, and rock mass integrity analysis. The specific workflow is as follows (as shown in Fig. 1).

Data collection

To clearly describe the method proposed in this paper, a synthetic combined point cloud standard dataset (Case A) and two actual rock slope datasets, namely a highway cut slope near Kingston, Canada (Case B) and a rural road cut slope in Yingshan, China (Case C), are selected as case studies, as shown in Fig. 2. Case A features a unique composition where the combined point cloud consists of multiple planes varying in size and orientation, with an average point cloud spacing of 1.00 m and a total of 263,919 points. In Case B, the primary lithology is quartz sandstone, and the point

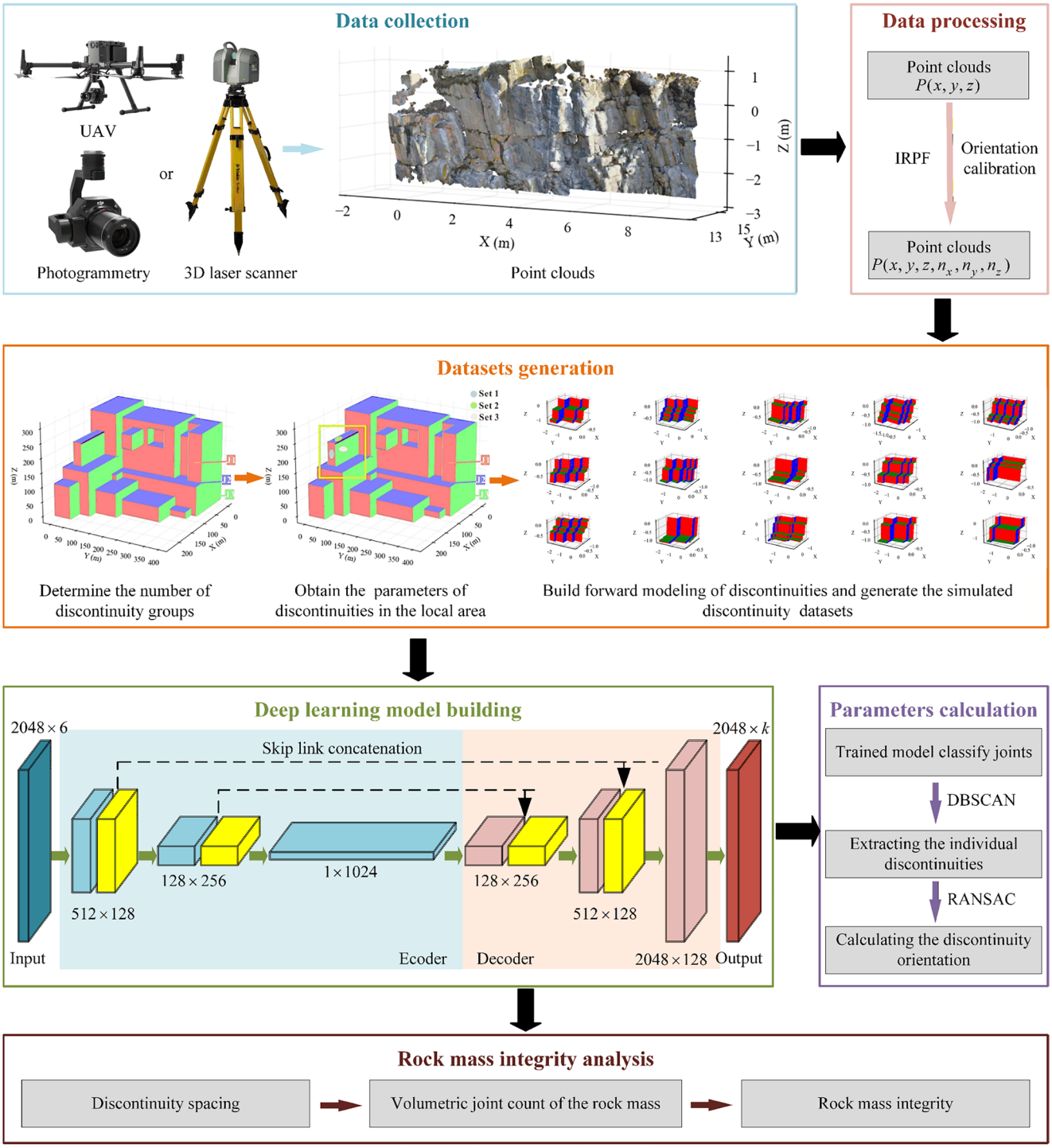


Fig. 1 Workflow for automatic identification of rock mass discontinuities and rock mass integrity analysis using the proposed Attention-PointNet++framework

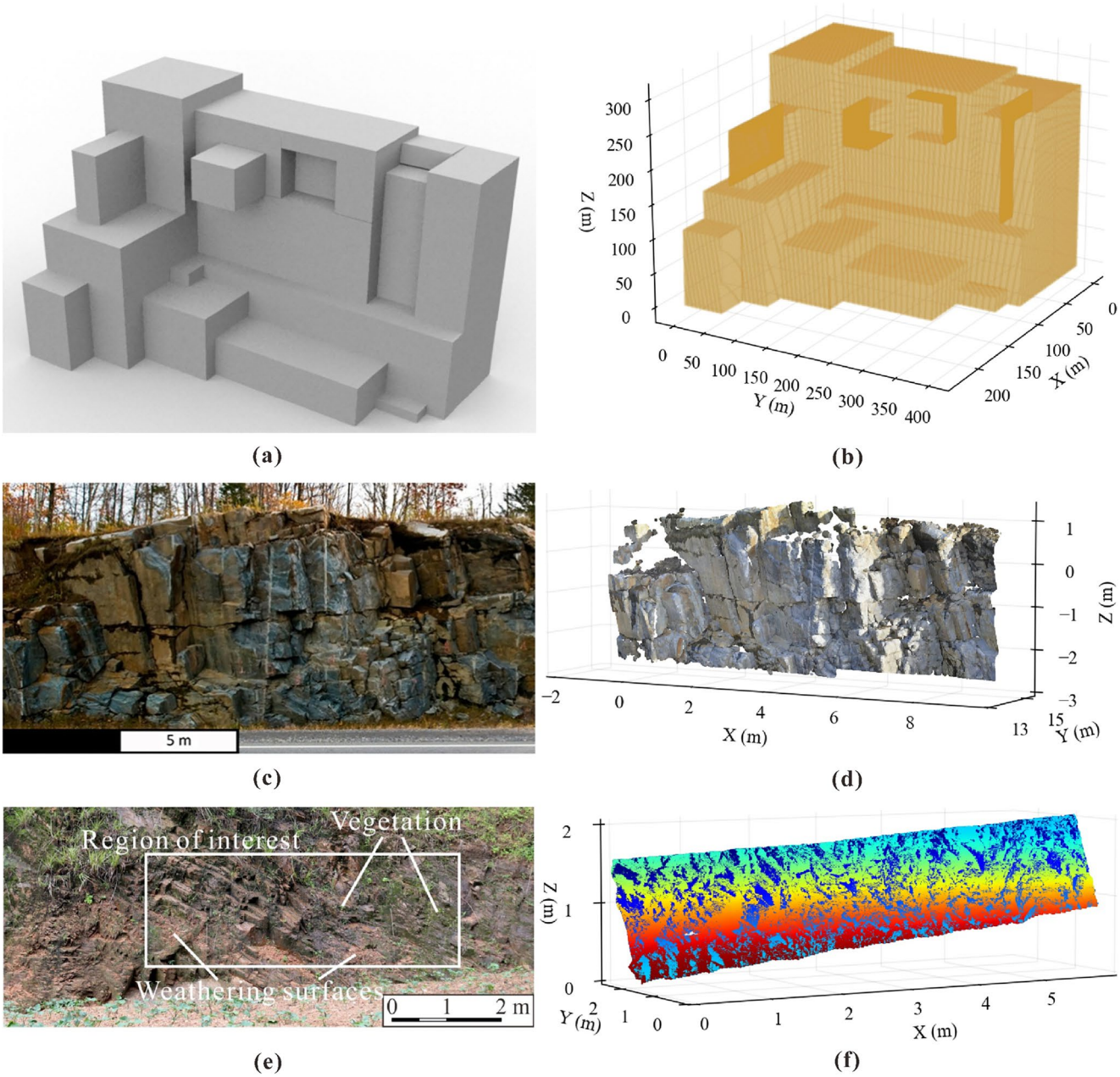


Fig. 2 Information of the three cases. a Photo of Case A. b Point cloud data of Case (A) c Photo of Case (B) d Point cloud data of Case B. e Photo of Case (C) f Point cloud data of Case C

cloud was shared by Lato et al. (2009). The data was collected using an Optech ILRIS-3D terrestrial laser scanner, and the region of interest in this study measures 11.0 m × 2.3 m × 4.1 m, encompassing 627,292 points. As for Case C, the main lithology is gneiss, and its point cloud was shared by Chen et al. (2024). The data acquisition was carried out with an Optech Polaris LR TLS terrestrial laser scanner. The region of interest in this research is 6.0 m × 2.4 m × 2.0 m, which contains 1,024,671 points.

Data processing

Calculate the point normal vectors

The surface of rock mass outcrops usually exhibits the coupled characteristics of macroscopic morphological undulations and microscopic roughness, involving multi-scale structures ranging from the millimeter scale to the meter scale. Such a complex surface is manifested as high-density

geometric discontinuous regions in the point cloud data, such as sharp features like the edges where different surfaces are adjacent, vertices, and the transition zones of curved surfaces. Aiming at the difficult problem of normal vector estimation for such complex point clouds, this paper introduces the Iterative Reweighted Plane Fitting (IRPF) method proposed by Wang et al. (2013). The core of this method lies in performing weighted plane fitting on the neighboring points of the target point x_i through an iterative optimization strategy. As shown in Fig. 3, this method comprehensively considers three key weight factors, namely distance weight, residual weight, and normal difference weight, to dynamically adjust the contribution of each neighboring point, thereby effectively suppressing noise, eliminating the interference of outliers, and accurately capturing the geometric information of sharp features in complex point cloud data.

The iteratively reweighted fitted plane (p) can be expressed as (Wang et al. 2013):

$$\begin{aligned} &p^t\left(d^t, \overrightarrow{\boldsymbol{n}}^t\right) = argmin\sum_{j=1}^{k} r_j^t w_d \\ &\left(x_j\right) w_r\left(r_j^{t-1}\right) w_n\left(\overrightarrow{\boldsymbol{n}}_j^{t-1}\right) ?\overrightarrow{\boldsymbol{n}}^t ? = 1 \end{aligned} \tag{1}$$

Where p^t denotes the iteratively reweighted fitting plane at the t-th iteration, d^t represents the distance from x_i to the fitting plane at the t-th iteration, $\overrightarrow{\boldsymbol{n}}^t$ is the normal vector of the fitting plane, which is also the normal vector of x_i, at the t-th iteration, x_j refers to the j-th neighboring point of x_i, r_j^t denotes the fitting residual of x_j at the t-th iteration, $w_d\left(x_j\right)$, $w_r\left(r_j^{t-1}\right)$, and $w_n\left(\overrightarrow{\boldsymbol{n}}_j^{t-1}\right)$ represent the distance weight, residual weight, and normal difference weight for x_j, respectively.

By dynamically adjusting these weights, the objective function is transformed into a positive semi-definite covariance matrix CM. The eigenvector corresponding to the minimum eigenvalue of CM is taken as the optimal normal vector of x_i:

$$CM = \begin{bmatrix} x_1 - x_i - u \\ x_2 - x_i - u \\ \cdots \\ x_k - x_i - u \end{bmatrix}^T \begin{bmatrix} x_1 - x_i - u \\ x_2 - x_i - u \\ \cdots \\ x_k - x_i - u \end{bmatrix} \tag{2}$$

$$u = \frac{\sum w_d(x_j) w_r(r_j) w_n\left(\overrightarrow{\boldsymbol{n}}_j\right)\left(x_j - x\right)}{\sum w_d(x) w_r(r_j) w_n\left(\overrightarrow{\boldsymbol{n}}_j\right)} \tag{3}$$

Calibration of the normal vector direction

Due to the rough and uneven surface of the rock mass, when performing local plane fitting on the neighborhood point cloud of the same discontinuity, the uneven distribution of residuals will cause a 180° reversal of the normal vector direction. This phenomenon is particularly prominent in the identification of discontinuities, causing subsequent classification algorithms to misjudge the orientation of discontinuities and leading to deviations in the stability analysis of the rock mass.

To address the above issues, a method for calibrating the direction of normal vectors based on the direction projection criterion can be adopted, as shown in Fig. 4. The consistency optimization of normal vectors is achieved through geometric projection constraints. The core lies in establishing a direction consistency criterion between the global reference direction and the local normal vectors. By projecting the normal vector $\overrightarrow{\boldsymbol{n}}$ onto the global reference direction $\overrightarrow{\boldsymbol{g}}$,

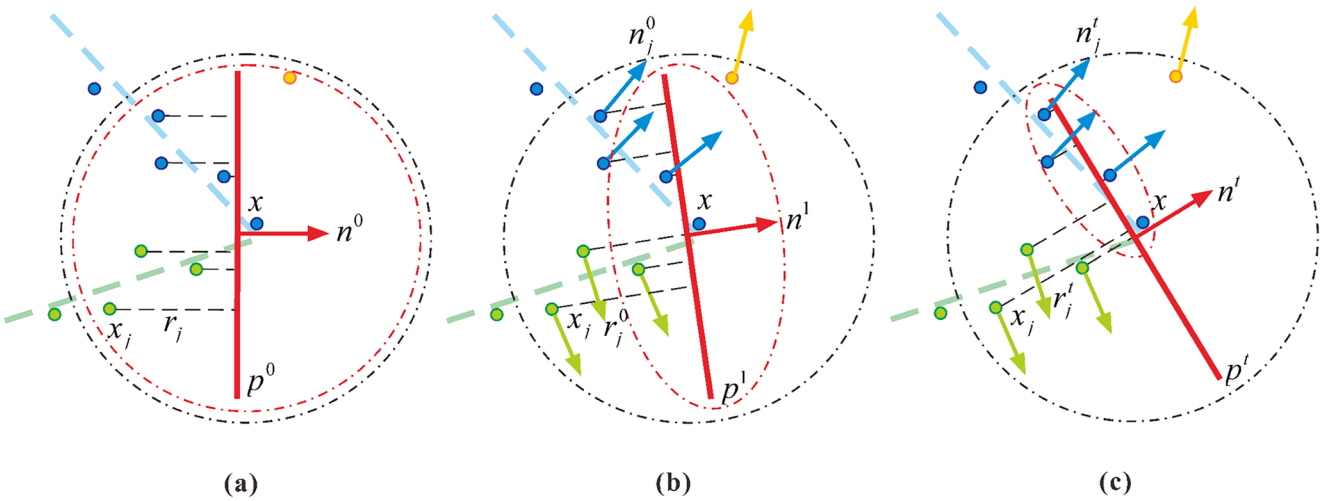


Fig. 3 Iterative reweighted plane fitting for points around sharp features, including distance, residual, and normal difference weights. a Initial stage; b Iterative process; c Iteration termination. (redrawn from Fig. 2 of Wang (Wang et al. 2013)

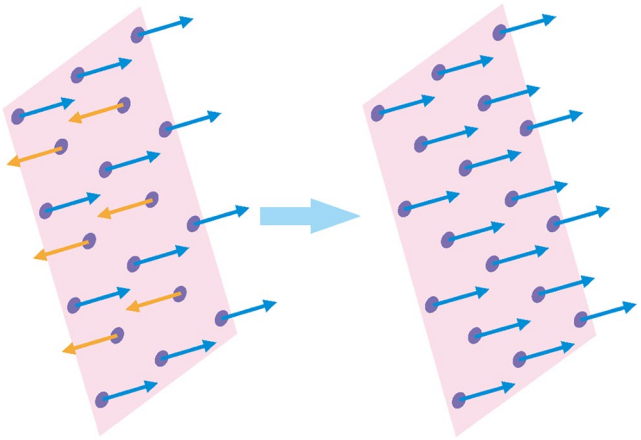


Fig. 4 Adjustment of the point normal vectors

the sign of the projection component s is used to determine the consistency between the direction of the normal vector and the global trend. The calculation formula for the projection component is:

$$s = \frac{\vec{n} \times \vec{g}}{|\vec{g}|} \tag{4}$$

In this context, when $s \geq 0$, the direction of the normal vector stays the same; when $s < 0$, the direction of the normal vector is reversed.

Generation of datasets

Determine the number of discontinuities groups

The normal vector field of point cloud data contains abundant geometric feature information, and the distribution of its directions directly reflects the spatial distribution pattern of the discontinuities in the rock mass. Visualizing the normal vectors using the HSV color space can transform abstract geometric attributes into color expressions that are in line with human visual perception.

Since the discontinuities in the same group possess similar geometric shapes in the spatial dimension, the normal vectors of the surface points also show remarkable consistency. When the normal vectors are mapped to the HSV color space according to specific rules, this consistency is intuitively manifested as color similarity. Based on this characteristic, by conducting a systematic statistical analysis of color categories, the number of discontinuities groups can be determined, as shown in Fig. 5.

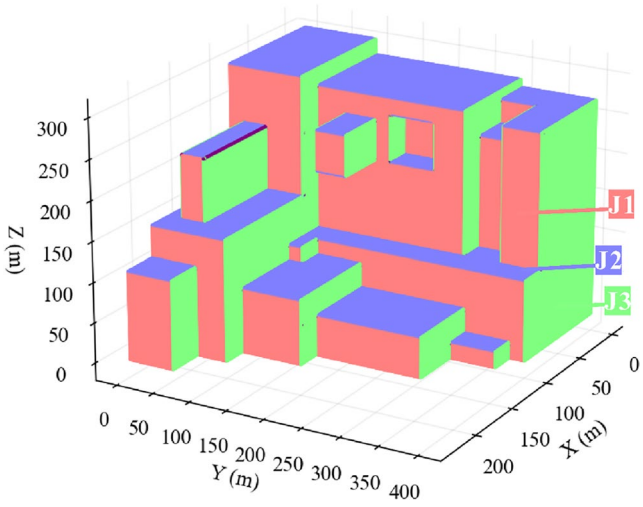


Fig. 5 Combined point cloud data. Map the normal vectors to the HSV color space. Different colors represent different discontinuities groups

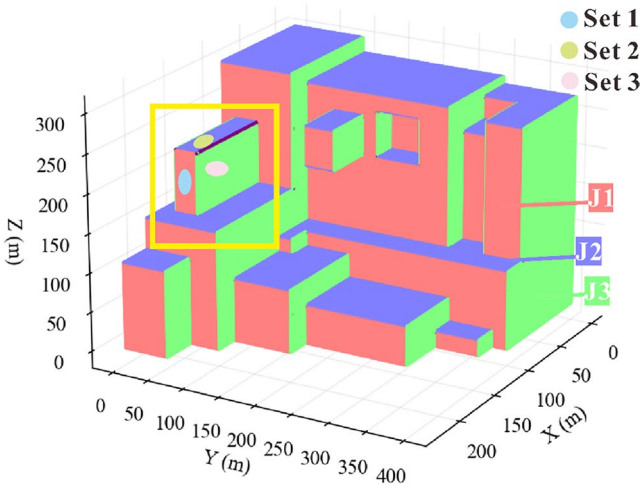


Fig. 6 Selection of regions from the combined point cloud data. The yellow boxes represent the selected local representative regions. The light blue ellipse indicates the region selected for the first discontinuities group, the tea-green ellipse represents the region selected for the second discontinuities group, and the pink ellipse shows the region selected for the third discontinuities group

Obtain the parameters of discontinuities in the local area

After mapping the normal vectors of the point cloud to the HSV color space, potential discontinuities groups can be automatically identified. Subsequently, typical representative regions are randomly selected from each discontinuities group, as shown in Fig. 6. The weighted average of the normal vectors of the point cloud within these regions for each discontinuities group is calculated as the normal vector

Table 1 The values of and

n_{pz}	K_1	Q_1
1	0	90
>0	1	0
0	0	0

Table 2 The values of and

n_{px}	n_{py}	K_2	Q_2
0	1	0	0
>0	>0	1	0
1	0	0	90
>0	<0	1	180
0	-1	0	180
<0	<0	1	180
-1	0	0	270
<0	>0	1	360
0	0	0	0

of the curved surface within that region, and then the normal vectors are converted into orientation parameters.

The dip angle and dip direction of the discontinuities can be calculated according to the following formulas:

$$\alpha = K_1 \times arccos\left(n_{pz}\right) + Q_1 \tag{5}$$

Where K_1 is the proportionality coefficient when calculating the dip angle, and Q_1 is the offset parameter when calculating the dip angle. The values are shown in Table 1.

$$\beta = K_2 \times \arctan\left(\frac{n_{px}}{n_{py}}\right) + Q_2 \tag{6}$$

Where K_2 is the proportionality coefficient when calculating the dip direction, and Q_2 is the offset parameter when calculating the dip direction. The values are shown in Table 2.

Build forward modeling

Set the parameter perturbation intervals centered around the orientation obtained above. The dip direction floating interval is $\alpha\ E\ [\alpha_0 - \Delta D, \alpha_0 + \Delta D] \cap [0°, 360°]$, where $\Delta D = 4°$. The dip angle floating interval is $\beta\ E\ [\beta_0 - \Delta A, \beta_0 + \Delta A] \cap [0°, 90°]$, where $\Delta A = 2°$. Generate $N = 100$ groups of random orientation combinations within the allowable parameter space. For each combination, the corresponding dataset can be generated in the following way.

As shown in Fig. 7, taking three groups of discontinuities as an example, first convert the orientations (dip angles and dip directions) of the three groups of discontinuities into normal vectors:

$$\vec{\boldsymbol{n}} = (sin\alpha sin\beta, sin\alpha cos\beta, cos\alpha) \tag{7}$$

Calculate the intersection line vectors of every two discontinuities: $\vec{l}_1 = \vec{n}_1 \times \vec{n}_2$, $\vec{l}_2 = \vec{n}_1 \times \vec{n}_3$, $\vec{l}_3 = \vec{n}_2 \times \vec{n}_3$.

Figure 8 shows the process of generating a simulated discontinuities dataset based on the forward modeling of discontinuities. Different colors are used to represent different discontinuities groups. The specific implementation idea is as follows:

(a.):Determine the initial plane J_1 using vectors $\vec{l}_1$ and $\vec{l}_2$. This plane is a parallelogram, with adjacent sides along $\vec{l}_1$ and $\vec{l}_2$ respectively. Set point cloud parameters: length *m*, width *n*, and unit spacing *d*. Generate point clouds column by column to get the point cloud data (red point set) of the initial discontinuity J_1, then

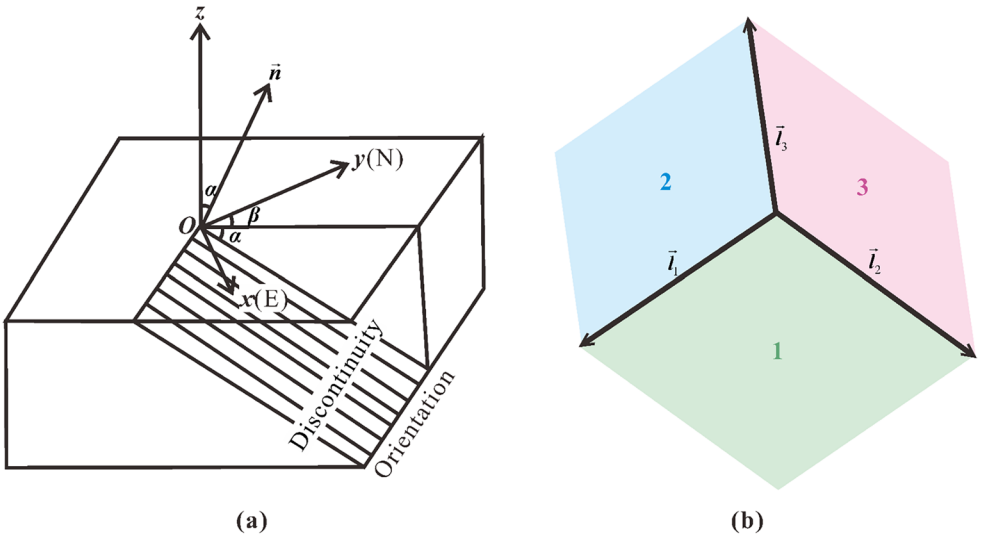


Fig. 7 The orientation and vector information of discontinuities. **a** The orientation of discontinuity and the normal vector. **b** The intersection line vectors of three groups of discontinuities

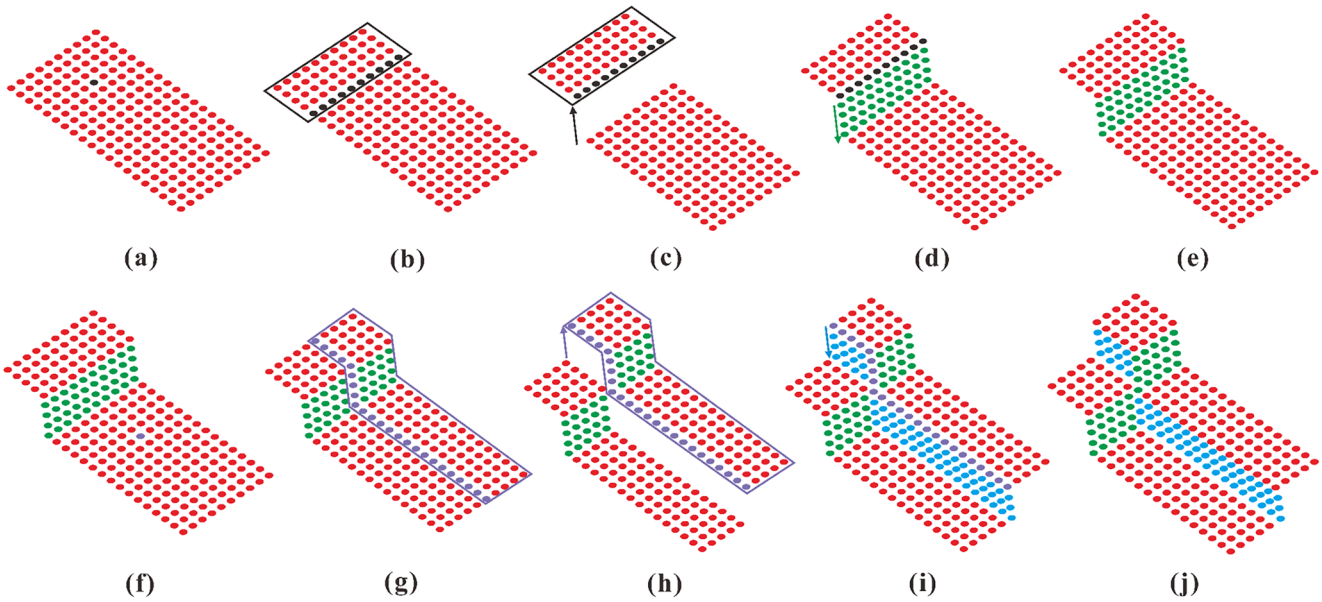


Fig. 8 Generation process of the simulated discontinuities dataset. The red point set represents the first group of discontinuities, the green point set represents the second group of discontinuities, and the blue point set represents the third group of discontinuities

store the point indices, select one point, and record its index i_1 black point).

(b.): Find the column of this point according to the index, and record the column and its left-side points (black box).

(c.):Move these points along $\vec{l}_3$ by d_1 to create a blank area.

(d-e): Find boundary points in the moved set, then fill new points column by column at spacing d along the direction opposite to $\vec{l}_3$—these points form discontinuity J_2 (green point set), resulting in two discontinuity groups.

(f-j): Select another point (purple point) from the current dataset, record its index i_2; locate its row, record the row and upper points (purple box); move them along $\vec{l}_3$ by d_2 to create a blank area; find boundary points, fill new points row by row at spacing d along the direction opposite to $\vec{l}_3$—these points form discontinuity J_3 (blue point set), finally getting three groups (J_1, J_2, J_3)that meet the given orientations.

Generate the simulated discontinuity datasets

Based on the obtained orientation parameters, as well as the geometric parameters of the discontinuities such as the set range parameters (*m* and *n*), spacing parameters (d_1 and d_2), and density parameter ($1/d$), a corresponding simulated point cloud dataset of discontinuities with definite labels can be generated using the established forward modeling of discontinuities, as shown in Fig. 9. In this study, a point cloud dataset containing 100 sets of simulated discontinuities was constructed. Among them, 80 sets (80%) were randomly sampled to form the training set, and the remaining 20 sets (20%) were used as the test set.

Deep learning model building

Network architecture

PointNet++ enables direct processing of unstructured point clouds via hierarchical feature learning, extracting local geometric features and gradually generating high-level semantic features through multi-level abstraction (Qi et al. 2017). However, it relies on a fixed geometric neighborhood relationship and a linear feature aggregation mechanism, and faces the bottleneck of insufficient local feature representation ability in complex spatial distribution scenarios. In order to break through this limitation, this study proposes an improved Attention-PointNet++network by embedding a self-attention mechanism into the intermediate layers of PointNet++. Specifically, a learnable dynamic feature correlation matrix is constructed to replace the traditional static neighborhood division, enabling the model to adaptively focus on critical geometric-semantic features, facilitate global feature interaction and information aggregation, and thus enhance local neighborhood feature extraction.

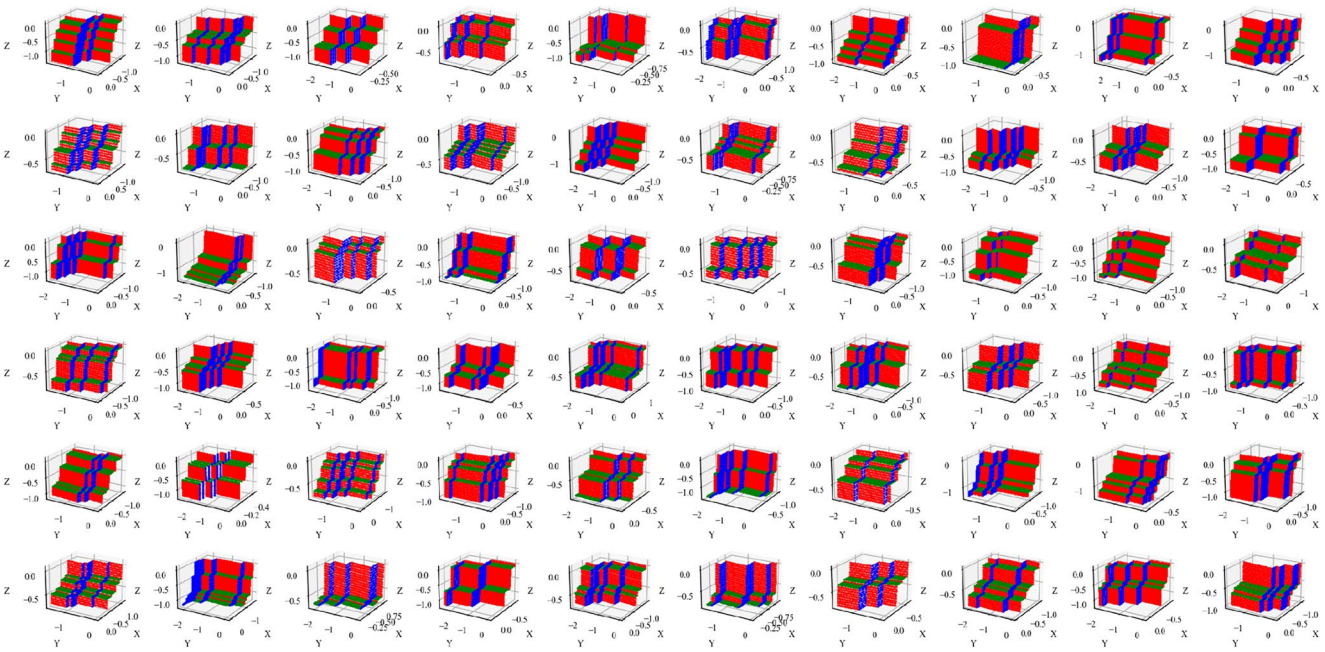


Fig. 9 Dataset generated based on the forward modeling. Different colors represent different discontinuities groups

The self-attention mechanism (Vaswani et al. 2017) maps input features to query (Q), key (K), and value (V) vectors, computes feature similarity via Q-K dot product, normalizes it with softmax to generate attention weights, and aggregates V to produce contextual features:

$$A(Q,K,V) = F_{\text{softmax}}\left(\frac{Q \bullet K^T}{\sqrt{d_k}}\right) V \tag{8}$$

Where d_k represents the dimension of the vector K.

To enrich feature representation, a multi-head attention mechanism is adopted, which computes multiple parallel self-attention heads and concatenates their outputs with a linear projection, capturing multi-perspective feature dependencies and further boosting point cloud feature learning. The specific implementation is as follows:

$$h_i = A\left(QW^{Q_i}, KW^{K_i}, VW^{V_i}\right) \tag{9}$$

$$F_{\text{multihead}}(Q,K,V) = C_{\text{concat}}(h_1, h_2, \ldots, h_n) W^o \tag{10}$$

Where $W^{Q_i} \in R^{D \bullet d_q}$, $W^{K_i} \in R^{D \bullet d_k}$, and $W^{V_i} \in R^{D \bullet d_v}$ are three groups of different weight matrices for Q, K, and V respectively; D represents the dimension of the word vector; d_v represents the dimension of the vector V; h_i represents the output vector of each head, where$1 \le i \le n$; W^o represents the projection weight; and n represents the number of heads.

The overall architecture of Attention-PointNet++is shown in Fig. 10:

Network environment and parameter settings

The Pytorch-CUDA 1.7.0 deep learning framework is adopted. The experimental environment is based on the Windows 11 operating system, with an NVIDIA GeForce RTX 3050 Laptop GPU and 4GB of running memory. The network parameter settings are as follows: the batch size is 8, the initial learning rate is 0.001, the number of point clouds in each batch is 2048, the maximum number of training epochs is 200, the optimizer is Adam, and the loss function is the cross-entropy loss function.

Network metrics

To quantitatively evaluate the performance of the deep learning network, this study uses the overall accuracy (OA) as the evaluation index. The overall accuracy OA is calculated as the ratio of the number of correctly

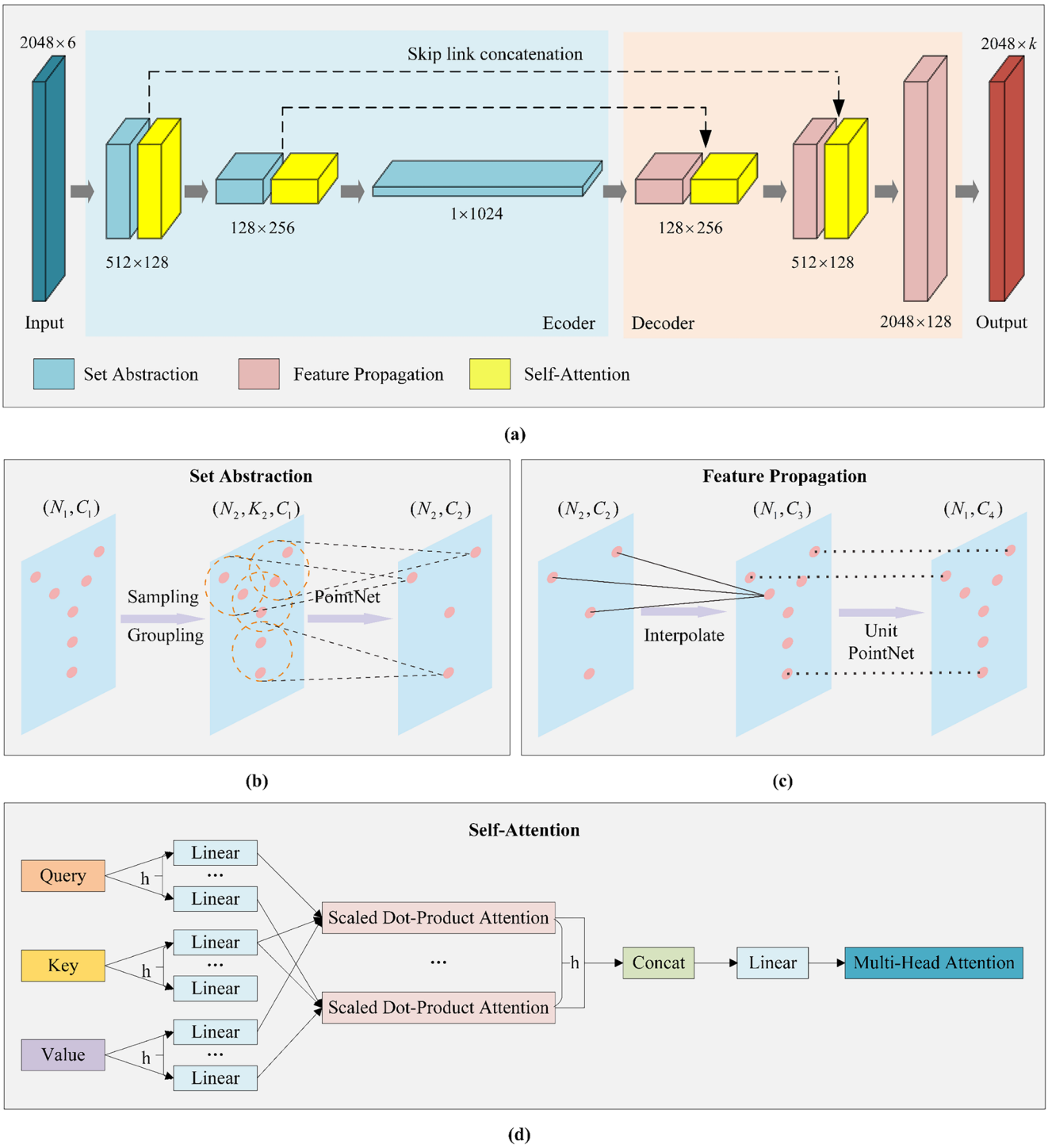


Fig. 10 Network architecture. **a** Overall network architecture. **b** Hierarchical feature extraction module. **c** Feature propagation module. **d** Self-attention mechanism module

classified points to the total number of points. The calculation formula is as follows:

$$OA = \frac{\sum_{i=0}^{k} x_{ii}}{\sum_{i=0}^{k}\sum_{j=0}^{k} x_{ij}} \tag{11}$$

Where k represents the number of categories of point clouds in the dataset; x_{ii} represents the number of point clouds that are accurately predicted; x_{ij} represents the number of point clouds whose label category is j but are predicted as category i.

Individual discontinuity identification and calculation of parameters

Extracting the individual discontinuities by DBSCAN

In this study, the focus of discontinuity identification is on the precise determination of individual discontinuities within each group of discontinuities, which can be achieved through clustering algorithms. DBSCAN, a well-known density-based clustering algorithm, has significant advantages in extracting individual discontinuities from a set of discontinuities. When running this algorithm, two key parameters need to be input: the search radius ϵ and the minimum number of points m_{pts}, which are used to establish a density threshold for distinguishing discrete points. However, it should be noted that fluctuations in the density of the dataset can affect the stability of the DBSCAN algorithm. To address this challenge, this study adopts a targeted strategy. By calculating the average spacing R of the input point cloud data and correlating the search radius ϵ with it (e.g., $\epsilon = 3R$, which can be adjusted according to the actual situation), the adaptability of the algorithm to datasets with different densities can be improved. Meanwhile, the parameter *minDis* is introduced, and extremely small clusters in the extraction results are regarded as noise and eliminated, thereby further enhancing the accuracy of discontinuity identification.

Calculating the discontinuity orientation by RANSAC

The orientation, as the core parameter characterizing the spatial geometric form of a discontinuity, consists of two elements: dip angle and dip direction. It plays an irreplaceable role in the stability analysis of rock mass joints (Sturzenegger and Stead 2009). Specifically, this parameter is not only a fundamental input variable for the block theory (Goodman 1995) and the rock mass classification system (Pantelidis 2009), but also a crucial basis for revealing the coupling relationship between the discontinuity and the direction of engineering loads. Therefore, the orientation is chosen to characterize the discontinuity. In this study, the RANSAC algorithm is adopted. Through iterative random sampling and model evaluation, it realizes noise-resistant plane fitting to determine the normal vector. Then, the dip angle and dip direction of the discontinuity are calculated according to Eqs. (5) and (6).

Rock mass integrity analysis

Rock mass integrity is a key factor affecting rock mass quality and the stability of rock slopes, and its core is reflected in two dimensions: geometric characteristics and physical and mechanical properties (Aydan et al. 1997; Xing et al. 2018; Wu et al. 2023). In the quantitative evaluation of rock mass integrity, commonly used indicators mainly include the average spacing of discontinuities (h) and the volumetric joint count of the rock mass (J_v), among others. Among these, conducting a qualitative analysis of rock mass integrity based on J_v can provide important support for assessing the stability state of rock slopes. Accordingly, this section will first perform quantitative calculations of discontinuity spacing and J_v based on the identified rock mass discontinuities, and then conduct a qualitative analysis of rock mass integrity based on the calculation results.

Discontinuity spacing

Discontinuity spacing is defined as the normal distance between discontinuity clusters within the same set. In natural rock masses, although the discontinuity clusters within the same set have similar orientations, they are not strictly parallel. For this reason, this study first adjusts the orientation of each cluster to the average orientation of the set to make them mutually parallel, and then uses the following formula to calculate the normal distance s_{ij} between any two discontinuity clusters i and j (Riquelme et al. 2015):

$$s_{ij} = \frac{|D_i - D_j|}{\sqrt{A^2 + B_2 + C^2}} \tag{12}$$

Where the plane equations of discontinuity clusters i and j can be expressed as:

$$\begin{cases} Discontinuityi : Ax + By + Cz + D_i = 0 \\ Discontinuityj : Ax + By + Cz + D_j = 0 \end{cases} \tag{13}$$

For a set containing m discontinuity clusters, the average normal spacing h can be obtained by averaging the normal distances between adjacent clusters, and the calculation formula is as follows:

Where $s_{i,i+1}$ represents the normal distance between the i-th discontinuity cluster and the $(i+1)$-th discontinuity cluster.

Volumetric joint count of the rock mass

The volumetric joint count of the rock mass (J_v) is defined as the number of discontinuities per cubic meter of rock mass. When the joints in each set are well-developed, the calculation formula can be expressed as (Sonmez and Ulusay 1999):

$$J_v = \sum_{k=1}^{n} \frac{1}{h_k} \# \tag{15}$$

Table 3 Correlation between, and the integrity of rock mass

J_v	< 3	3~10	10~20	20~35	≥ 35
K_v	> 0.75	0.55~0.75	0.35~0.55	0.15~0.35	≤ 0.15
Rock mass integrity	Intact	Less intact	Less fractured	Fractured	Very fractured

Where h_k is the average spacing of discontinuities in the k-th sets.

Based on J_v of the rock mass, a qualitative classification of rock mass integrity can be further carried out. The corresponding relationships between J_v, the rock mass integrity coefficient K_v, and the degree of rock mass integrity are shown in Table 3:

$$h = \frac{1}{m-1}\sum_{i=1}^{m-1} s_{i,i+1} \tag{14}$$

Case study

Case A

Judging from the identification results of the combined point cloud standard dataset shown in Fig. 11, the model successfully identified three groups of discontinuities. The DBSCAN algorithm was used to identify individual discontinuities, and a total of 45 discontinuities were identified and visually distinguished by different colors. Specifically, there are 15 independent discontinuities in each of Group J_1, Group J_2, and Group J_3.

In the combined point cloud standard dataset, the orientation of the discontinuities can be accurately determined. This characteristic provides extremely favorable conditions for verifying the correctness of the method proposed in this study. Because we can judge the reliability of the method by calculating the deviation between the orientation results of the discontinuities obtained by the proposed method and the known true values. Through rigorous calculation, the dominant orientation data of the three groups of discontinuities shown in Table 4 were obtained. After an in-depth analysis of the identification results of the discontinuities in the combined point cloud dataset, it was found that the identification results obtained by using the method proposed in this study are highly reliable. In the test of the standard dataset, the identification accuracy of this method reached 100.00%, and this data strongly confirms the effectiveness and accuracy of the method.

Case B

The normal vectors of the point cloud data of the highway cutting slope near Kingston, Canada, were calculated, and the directional consistency calibration was carried out. Figure 12 shows the three-dimensional visualization results of the rock mass discontinuities based on the normal vector mapping in the HSV color space, clearly presenting the spatial distribution of the three groups of dominant discontinuities. In this study, the area enclosed by the yellow frame was selected for a refined analysis, and the specific implementation steps are as follows: A small area was randomly selected from each of the three groups of discontinuities in this area (the blue, green, and pink areas). The point cloud data of the selected

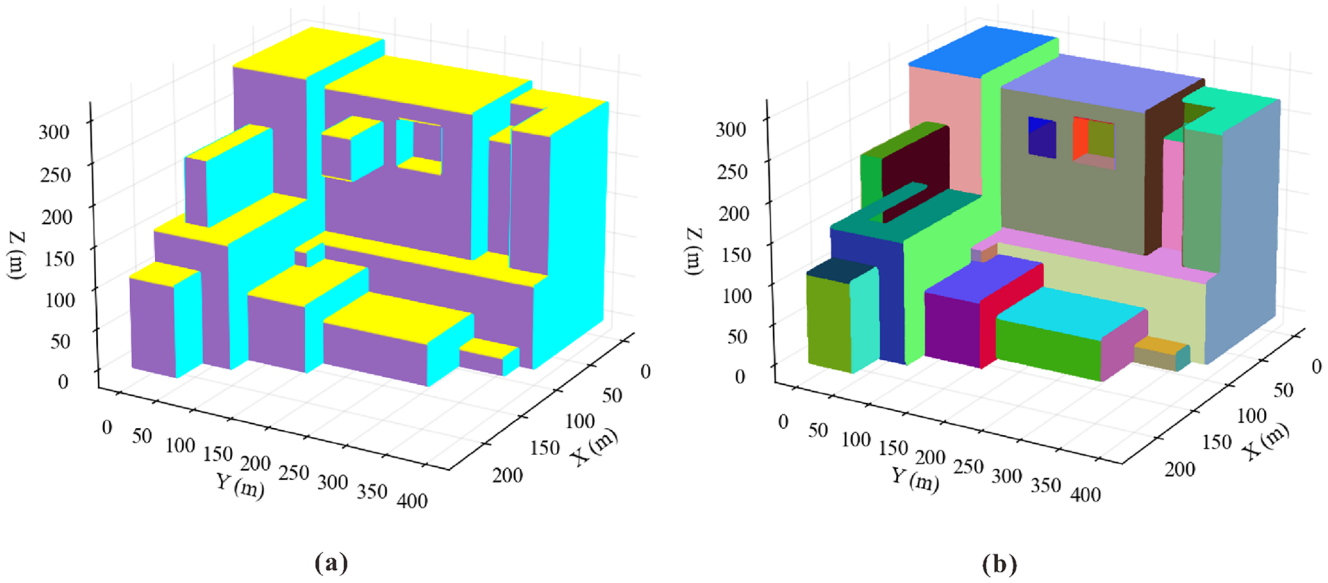


Fig. 11 Identification results of Case A. **a** Overall identification results of Case A. **b** Identification results of individual discontinuities in Case A

Table 4 Comparison between the identification results of the proposed method and the true values

Dataset	Combined point cloud	
	The proposed method	True value
Orientation /°	90/90	90/90
	00/00	00/00
	00/90	00/90
Orientation accuracy	**100.00%**	

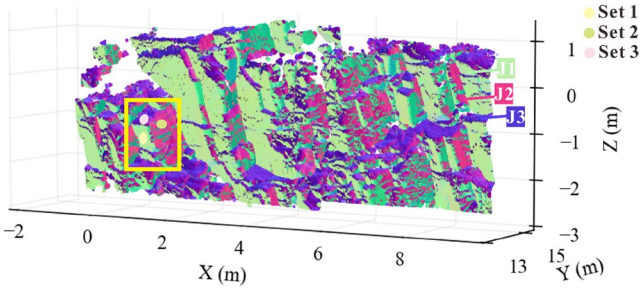


Fig. 12 Color mapping of normal vectors in Case B. The yellow frame indicates the selected local representative area. The light yellow ellipse represents the selected area of the first group of discontinuities, the tea-green ellipse represents the selected area of the second group of discontinuities, and the pink ellipse represents the selected area of the third group of discontinuities

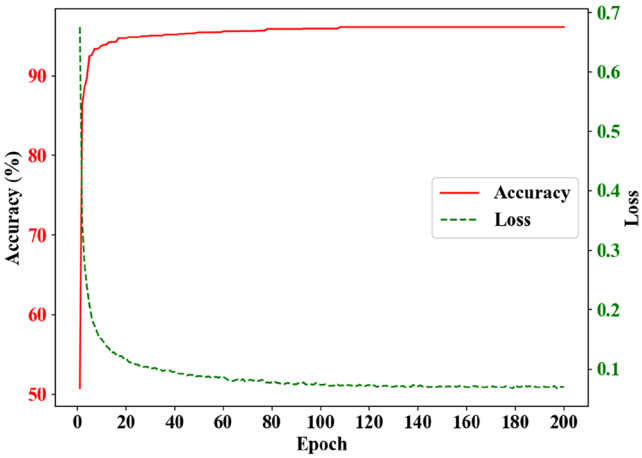


Fig. 13 Training accuracy values and training loss values obtained by using the Attention-PointNet++network for Case B

small areas were processed, their normal vectors were calculated and further converted into orientation data. Subsequently, taking the obtained orientation data as the core basis, the corresponding floating range and the number of simulated datasets were set. Among them, $\Delta D = 4°$ and $\Delta A = 2°$, $N = 100$. These discontinuity simulated datasets generated through specific settings will be used to train the improved Attention-PointNet++network model constructed in this study.

Figure 13 shows the accuracy and loss curves during the training of the deep learning network for Case B. It can be seen that the training accuracy gradually increases and tends to be stable, while the loss gradually decreases and also stabilizes. This indicates that the network model has reached the optimal state, and at this time, the overall accuracy is close to 96%. After the training is completed, the real point cloud data of Case B are input into the model for identification. According to the identification results, the overall accuracy OA is calculated, and the OA is 93.1%, which shows that the network in this paper has a good ability to identify rock mass discontinuities.

Figure 14a shows the identification results of the discontinuities in Case B. It can be clearly seen from the figure that different colors correspond to different discontinuity groups, and the three groups of discontinuities in the rock mass have been accurately extracted, which strongly proves the accuracy of this improved network in the task of discontinuity identification. Figure 14b shows the results of extracting individual discontinuities using the DBSCAN algorithm under specific parameter settings, that is, the search radius $\varepsilon = 3R$, the minimum number of points for clustering $m_{pts} = 5$, and the minimum number of points in a cluster $minDis = 100$. Under this parameter combination, the algorithm has successfully and effectively extracted individual discontinuities from the complex point cloud data. At the same time, it has accurately identified and removed some discontinuities with too small a scale and noise points, significantly improving the accuracy and clarity of the discontinuity extraction results, laying a good foundation for the subsequent in-depth analysis and research on the discontinuities.

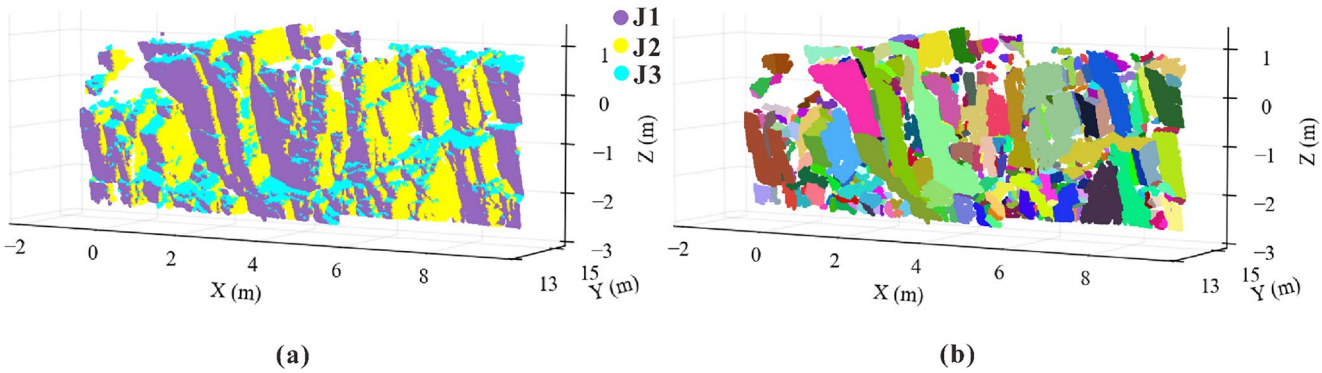


Fig. 14 Identification results of Case B. **a** Overall identification results of Case B. **b** Identification results of individual discontinuities in Case B

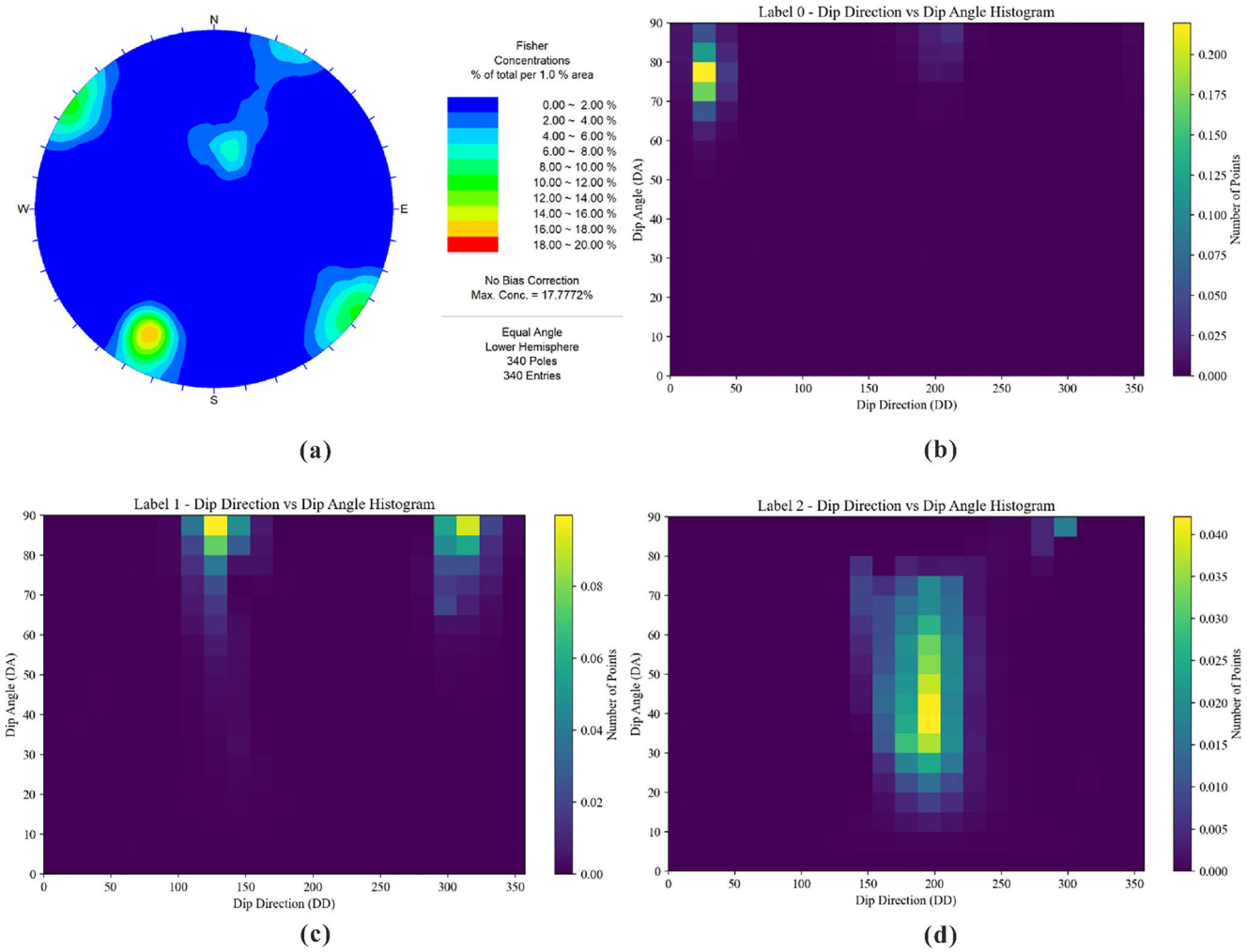


Fig. 15 The orientation information of discontinuities. **a** Stereographic projection of the orientations of the three groups of discontinuities in Case B**. b-d** 2D histograms of the orientations of the three groups of discontinuities in Case B

Through precise calculation, the dominant orientations of the three groups of discontinuities were obtained. To visually present the distribution characteristics of these discontinuity orientations, Fig. 15 uses two methods, stereographic projection and 2D histogram, to display the orientation distribution of each group of discontinuities, providing a visual representation of their overall orientation characteristics. The dominant orientations of the three groups of discontinuities are 28.2°/77.4°, 127.8°/87.5°, and 194.9°/37.6° respectively. From Fig. 15, the concentration tendency and dispersion degree of each group of discontinuities in spatial distribution can be clearly observed, which provides an important basis for in-depth analysis of rock mass structural characteristics and stability.

To further verify the accuracy of the results, manual measurements were first conducted on the 3D model to obtain the ground truth of discontinuity orientations, which served as the core validation benchmark. Meanwhile, CloudCompare software was employed as an auxiliary tool to measure

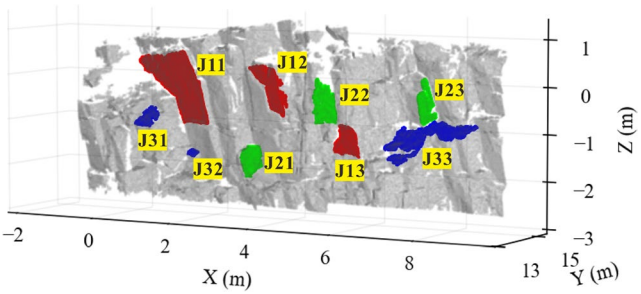


Fig. 16 Selected **clusters** for parameter calculation in Case B. Red represents the first group of discontinuities, green represents the second group of discontinuities, and blue represents the third group of discontinuities

the discontinuity orientations for comparative analysis in line with the common practice in the field. Figure 16 shows the nine clusters used for verification (three clusters were selected from each group of discontinuities). As indicated by the comparison results in Table 5, for these nine

Table 5 Comparison of orientations among CloudCompare, manual measurement, and the proposed method

Discontinuity set	Joint label	CloudCompare	Manual measurement	The proposed method	E1	E2
		DD/DA(°)	DD/DA(°)	DD/DA(°)	Δ|DD|/Δ|DA| (°)	Δ|DD|/Δ|DA| (°)
1	J11	28.9/76.7	29.8/76.3	29.6/76.8	0.7/0.1	0.2/0.5
1	J12	29.9/78.7	30.2/78.9	28.9/79.4	1.0/0.7	1.3/0.5
1	J13	28.1/79.8	27.2/79.1	26.6/79.8	1.5/0.0	0.6/0.7
2	J21	305.2/87.9	127.0/88.4	127.9/88.0	2.7/0.1	0.9/0.4
2	J22	131.8/87.5	132.9/85.8	133.6/86.0	1.8/1.5	0.7/0.2
2	J23	303.9/85.2	304.7/88.1	123.7/88.5	0.2/3.3	1.0/0.4
3	J31	196.2/39.1	194.1/38.6	194.8/40.1	1.4/1.0	0.7/1.5
3	J32	194.6/42.3	196.7/43.1	195.5/42.4	0.9/0.1	1.2/0.7
3	J33	190.7/41.4	188.7/44.0	188.5/44.2	2.2/2.8	0.2/0.2
Maximum deviation					2.7/3.3	1.3/1.5
Average deviation					1.4/1.1	0.8/0.6

* DD is the dip direction, DA is the dip angle, ΔDD is the dip direction deviation, and ΔDA is the dip angle deviation. E1 represents the absolute value of the error between the results obtained from the proposed method and those acquired using CloudCompare. E2 represents the absolute value of the error between the results obtained from the proposed method and manual measurement.

Table 6 Rock mass integrity analysis for Case B

Discontinuity set	Maximum spacing (m)	Minimum spacing (m)	Average spacing (m)	Volumetric joint count (joint count/m^3)	Rock mass integrity
1	0.338	0.001	0.125	0.192	Intact
2	0.799	0.006	0.169		
3	0.596	0.006	0.173		

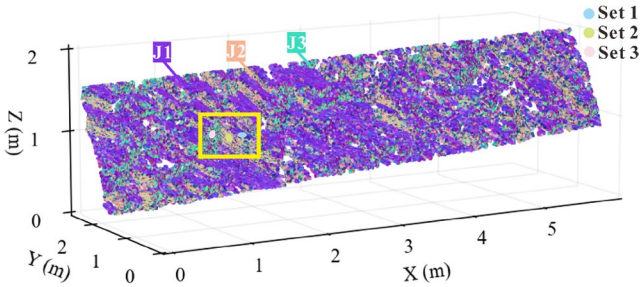


Fig. 17 Color mapping of normal vectors in Case C. The yellow frame indicates the selected local representative area. The light blue ellipse represents the selected area of the first group of discontinuities, the tea-green ellipse represents the selected area of the second group of discontinuities, and the pink ellipse represents the selected area of the third group of discontinuities

clusters, the average errors of dip direction and dip angle are 1.4°/1.1° and 0.8°/0.6°, respectively. This data fully demonstrates that the method proposed in this paper has excellent performance in the identification of discontinuity orientations, with extremely high accuracy and reliability, providing solid data support for related research and engineering applications.

To analyze the rock mass integrity of Case B, the discontinuity spacing and volumetric joint count were calculated separately, as shown in Table 6. It can be seen from the table that the rock mass of the highway cut slope near Kingston, Canada, is intact and exhibits high overall stability.

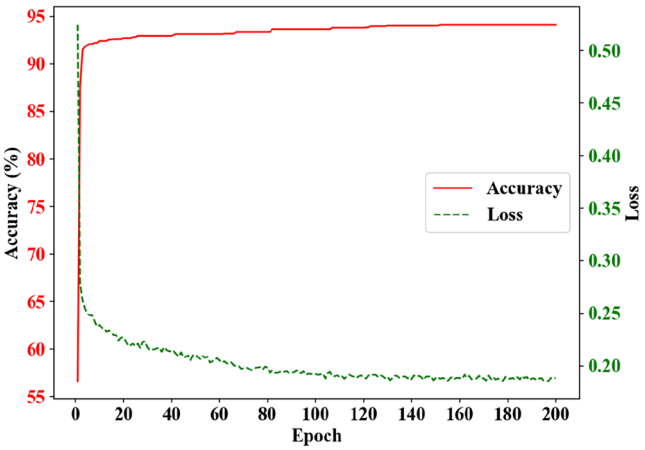


Fig. 18 Training accuracy values and training loss values obtained by using the Attention-PointNet++network for Case C

Case C

Similar to Case B, for Case C, after mapping the normal vectors in the HSV color space, a small area was randomly selected from each group of discontinuities within the area enclosed by the yellow frame, as shown in Fig. 17. The normal vectors were calculated and converted into orientation data. Taking the orientation data as the core, $\Delta D = 4°$, $\Delta A = 2°$, and $N = 100$ were set. Based on the forward modeling, a simulated dataset of discontinuities was generated and input into the Attention-PointNet++network model for training.

Figure 18 shows the accuracy and loss curves during the training of the deep learning network for Case C. When the network training converges, the overall accuracy is close to 94%. The real point cloud data of Case C are identified based on the trained model, and the OA calculated according to the identification results is 90.9%, indicating that the network proposed in this paper has good robustness.

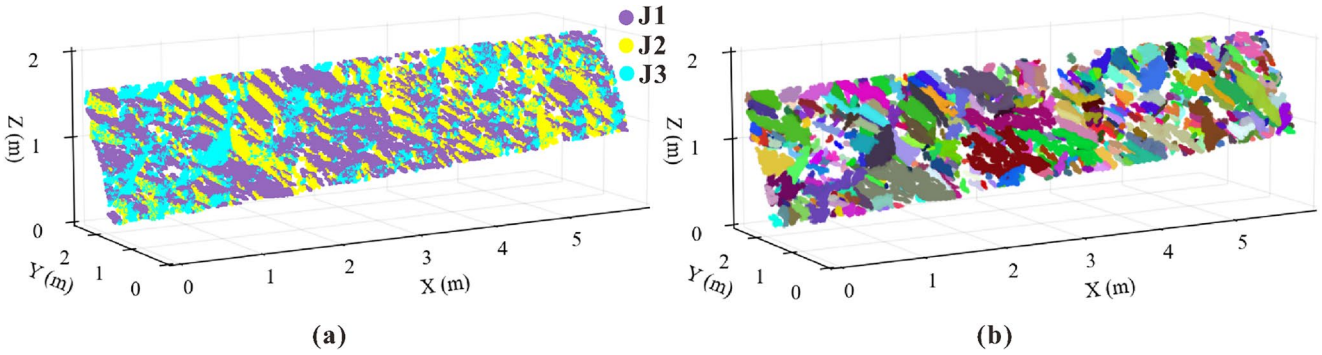


Fig. 19 Identification results of Case C. **a** Overall identification results of Case C. **b** Identification results of individual discontinuities in Case C

Figure 19a shows the identification results of the discontinuities in Case C, which intuitively indicates that the network has successfully and accurately extracted the three groups of discontinuities in the rock mass from the complex data, fully demonstrating the reliability of the improved network in the task of discontinuity identification. Figure 19b shows the results of extracting individual discontinuities using the DBSCAN algorithm under the specific parameter settings of $\varepsilon = 3R$, $m_{pts} = 5$, and $minDis = 100$.

To more intuitively present the distribution patterns of the discontinuity orientations, Fig. 20 visualizes the orientation distribution of each group of discontinuities by combining

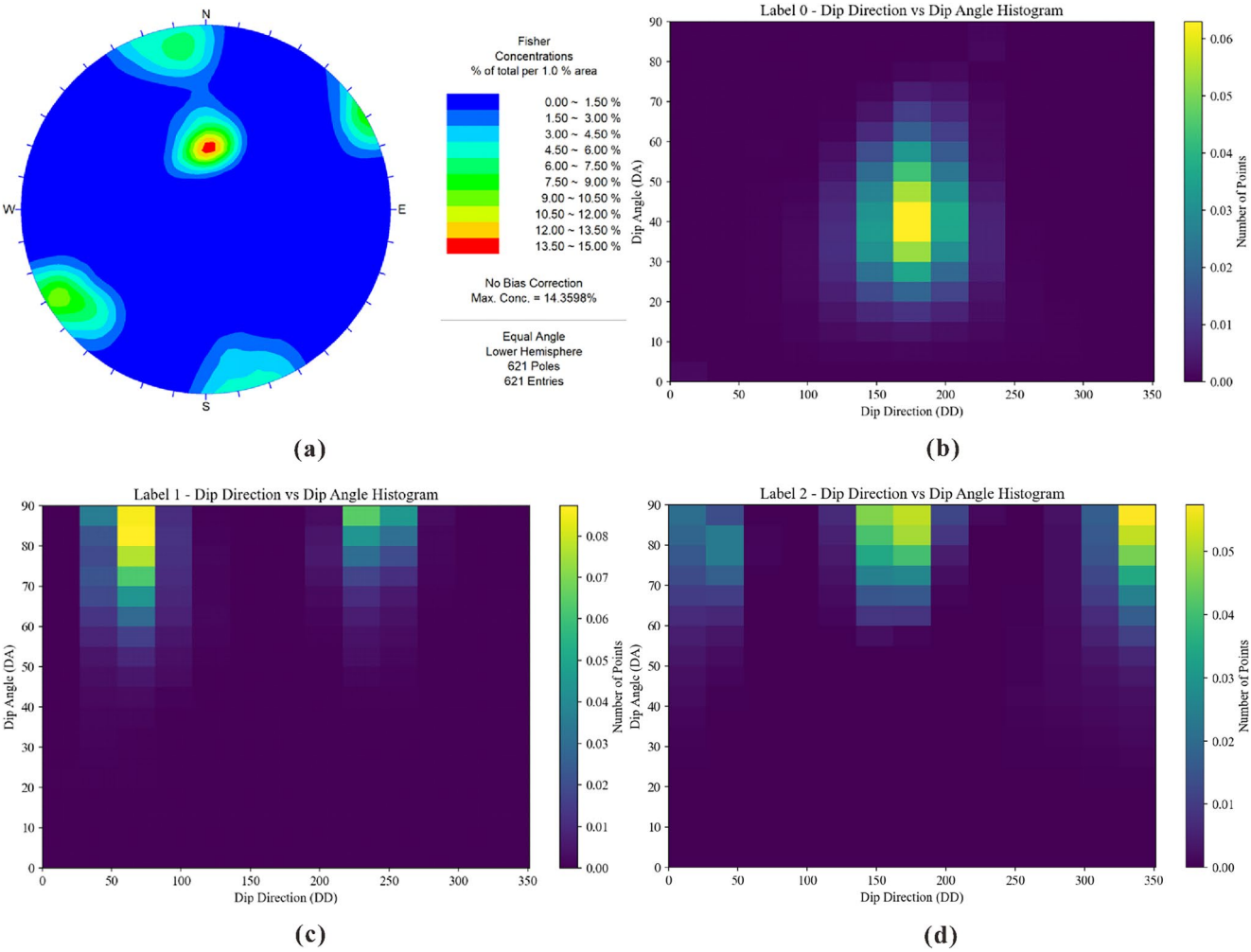


Fig. 20 The orientation information of discontinuities. **a** Stereographic projection of the orientations of the three groups of discontinuities in Case C. **b-d** 2D histograms of the orientations of the three groups of discontinuities in Case C

stereographic projection and 2D histogram. Through rigorous calculations, the dominant orientations of the three groups of discontinuities are 175.5°/ 37.5°, 65.0°/ 82.5°, and 338.4°/ 87.5° respectively.

To further ensure the reliability and accuracy of the results, the results obtained by the method proposed in this paper were compared with the results of Chen et al. (2024). Figure 21 shows the nine clusters used for the comparison. From the comparative data shown in Table 7, it can be seen that the average error in dip direction for these nine clusters is only 1.9°, and the average error in dip angle is 1.4°. Such small errors fully demonstrate the excellent performance of the method proposed in this paper. It can output high-precision and highly reliable data in the measurement of discontinuity orientations, providing strong technical support for the research on rock mass structures and related engineering applications.

* DD is the dip direction, DA is the dip angle, ΔDD is the dip direction deviation, and ΔDA is the dip angle deviation.

To analyze the rock mass integrity of Case C, the discontinuity spacing and volumetric joint count of the rock mass were calculated separately, as shown in Table 8. It can be seen from the table that the rock mass of the rural road cut slope in Yingshan, China, is fractured and exhibits poor overall stability.

Table 7 Comparison of orientations between the proposed method and the results of Chen et al. (2024)

Discontinuity set	Joint label	Chen et al. (2024)	The proposed method	Absolute error
		DD/DA(°)	DD/DA(°)	Δ|DD|/ Δ|DA| (°)
1	J11	171.5/35.6	172.7/35.3	1.2/0.3
1	J12	208.5/39.7	207.5/37.1	1.0/2.6
1	J13	195.8/31.4	197.8/30.9	2.0/0.5
2	J21	60.1/80.8	59.0/80.5	1.1/0.3
2	J22	60.1/84.3	57.4/86.9	2.7/2.6
2	J23	58.7/84.8	60.0/84.2	1.3/0.6
3	J31	340.2/87.2	339.0/82.9	1.2/4.3
3	J32	168.5/83.8	167.6/83.8	0.9/0.0
3	J33	351.5/84.9	356.8/86.2	5.3/1.3
Maximum deviation				5.3/4.3
Average deviation				1.9/1.4

Table 8 Rock mass integrity analysis for Case C

Discontinuity set	Maximum spacing (m)	Minimum spacing (m)	Average spacing (m)	Volumetric joint count (joint count/ m^3)	Rock mass integrity
1	0.252	0.001	0.121	25.530	Fractured
2	0.061	0.002	0.021		
3	0.018	0.003	0.009		

Discussion

Accuracy of different networks

In order to verify the identification accuracy of the discontinuities of the Attention-PointNet++network model, in this study, the same point cloud dataset of discontinuities was used and input into the classic PointNet++network and the Attention-PointNet++network respectively for experiments to test the effect of the improved network model in this study, as shown in Fig. 22. Moreover, the identification results of this study were compared with the identification results of the CNN model built by scholars such as Ge et al. (2025) using the GoogLeNet network and the calculated orientation parameters, so as to study the identification performance of these two network models for the same point cloud data of discontinuities.

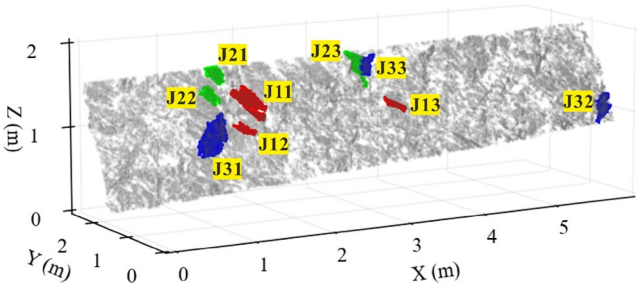


Fig. 21 Selected **clusters** for parameter calculation in Case C. Red represents the first group of discontinuities, green represents the second group of discontinuities, and blue represents the third group of discontinuities

As shown in the experimental results in Table 9, it was found that the identification results of the three groups of discontinuities by the Attention-PointNet++network model in this paper are significantly better than those of the classic PointNet++network. The overall accuracy is increased by 10% points compared with that of the classic PointNet++network. At the same time, due to the introduction of the self-attention mechanism, the network structure is relatively more complex, leading to a slight increase in its running time and memory requirements, including a small extension of training time and recognition time, as well as a slight rise in peak GPU memory usage and system RAM consumption. Overall, the slight increase in time and memory is negligible compared with its significant improvement in accuracy, making the Attention-PointNet++significantly outperform PointNet++in this task.

As shown in Table 10, for discontinuity set 1, our proposed Attention-PointNet++achieved 88.4% test-set accuracy, slightly lower than that of CNN at 90%. However, its calculated dip direction data were closer to the reference values of Lato et al. (2009), whereas the absolute error of CNN reached approximately 9°. For discontinuity sets 2 and 3, Attention-PointNet++outperformed CNN in both

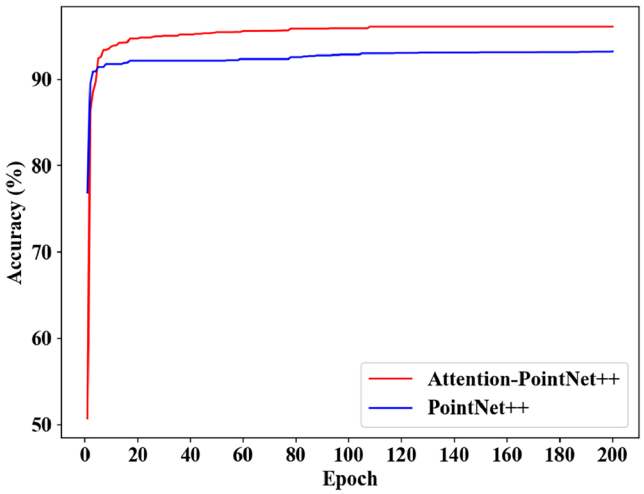


Fig. 22 Accuracy curves during the training of PointNet++and Attention-PointNet++for Case B

test-set accuracy and average orientation accuracy, with calculated data closer to Lato et al. (2009) and lower overall absolute errors. Across multiple discontinuity sets, our method excelled in average orientation accuracy, surpassing CNN in most test cases. This advantage stems from the attention mechanism, which enables the model to focus on key features, filter interference, and significantly improve discontinuity identification accuracy.

Limitations and potentials

The benchmark case adopted in this study is designed as a simple orthogonal planar structural network, and its core design logic has a high degree of consistency with the basic verification requirements in the process of dataset construction. In the research on rock mass discontinuity identification based on deep learning, the construction of datasets often relies on manual operations, which is prone to problems such as low efficiency and large errors. This study effectively solves this technical problem by generating training sets through discontinuity forward modeling technology. The benchmark case adopts a simple orthogonal structure, whose core purpose is to construct a standardized and low-interference verification scenario, eliminate the interference of complex structural parameters, accurately verify the reliability of the Attention-PointNet++algorithm, and lay a solid theoretical and practical foundation for subsequent algorithm verification in complex rock mass scenarios.

Table 10 Comparison of identification effects and orientations between Attention-PointNet++and CNN for Case B

Discontinuity set	Lato et al. (2009) (°)	Average orientation(°)		Test Set Accuracy(%)	
		Proposed method	Ge et al. (2025)	Proposed method	Ge et al. (2025)
1	194/34	194.9/37.6	186.2/32.5	88.4	90
2	29/76	28.2/77.4	26.3/86.7	94.2	88
3	309/90	127.5/87.5	319.0/87.3	93.9	92

Different from the simplified design of the benchmark case, the practical cases in this study are real rock mass discontinuity identification cases, although all containing three sets of discontinuities, are not simple orthogonal planes but non-orthogonal complex structures consistent with the actual geological environment. Discontinuities in natural rock masses are affected by natural factors such as geological evolution and weathering and erosion, coupled with the rock mass fragmentation effect caused by artificial excavation, making it difficult for the discontinuity sets of actual rock masses to exhibit strict orthogonal characteristics. Each set of discontinuities in the practical cases of this study has independent orientation parameters, with significant differences in dip direction and dip angle, showing non-orthogonal distribution with each other. In addition, they are accompanied by complex characteristics such as local fragmentation and uneven spacing, which can fully reflect the complexity of real rock mass structures and are not limited to simplified rock structure scenarios, further confirming the adaptability of the proposed algorithm in practical engineering.

Based on the real rock mass structural characteristics revealed by the previous practical cases, aiming at the defects of existing discontinuity forward modeling such as lack of microscopic characteristics and limited applicable scenarios, future research will be further promoted based on the existing optimization directions: further expanding to multiple sets of non-orthogonal and highly fragmented complex structural networks, optimizing the discontinuity forward modeling method by introducing more advanced algorithms for training set generation and integrating microscopic morphological parameters, significantly improving the versatility and adaptability of the algorithm, promoting its engineering application in complex rock mass engineering scenarios, and thus making up for the deficiencies of existing research.

Table 9 Comparison of identification resultsof PointNet++and Attention-PointNet++for Case B

Network	Accuracy (%)				Running time		Memory requirements (GB)	
	J1	J2	J3	OA	Training stage	Recognition stage(s)	Peak GPU memory usage	System RAM consumption
PointNet++	72.3	87.2	79.1	83.1	14min39s	2.6	3.15	1.45
Attention - PointNet++	88.4	94.2	93.9	93.1	15min11s	2.8	3.23	1.49

Conclusion

This study proposes an innovative method for identifying rock mass discontinuities by integrating the forward modeling of discontinuities with the Attention-PointNet++network model enhanced by the attention mechanism. The main research conclusions are summarized as follows: (1) The IRPF method introduced in this study, through the ingenious introduction of the iterative reweighting mechanism, can effectively suppress noise, eliminate the interference of outliers, and accurately capture the geometric information of sharp features when dealing with complex point cloud data. (2) The proposed method for generating a training set based on local area constraints effectively overcomes the limitations of the traditional manual point selection process, which is cumbersome, inefficient, and affected by the randomness of samples on classification accuracy, in the field of training point set construction. Through the innovative area constraint mechanism, the spatial correlation and feature representativeness of training samples are significantly improved. (3) The Attention-PointNet++network architecture constructed by integrating the self-attention mechanism and PointNet++achieves cross-scale interaction of point cloud features and global context awareness in the optimization of point cloud classification tasks. It effectively solves the problem of the lack of spatial dependence in the neighborhood feature aggregation process of the traditional PointNet++model and significantly improves the accuracy of discontinuity identification.

In conclusion, through the collaborative optimization of multiple modules, this study has formed a technical closed loop in three key aspects: point cloud data processing, training dataset construction, and deep learning network optimization. The newly proposed method has been applied to a standard dataset and two datasets of actual cut slope cases. The principle and process of the method were elaborated on the standard dataset, and its correctness was verified. In the two real cases, by comparing with previous results, the reliability and robustness of the results of this method were highlighted.

It is worth noting that due to the limitations of the forward modeling and the generation method of the training set, the applicable scope of this method in practical applications is limited. It has a relatively high degree of adaptability mainly in scenarios where the characteristics of the discontinuities are relatively uniform and the rock masses are mainly composed of fewer groups of discontinuities. Therefore, future research work will focus on optimizing the forward modeling and making efforts to improve the generation mechanism of the training set, so as to enhance the performance of the model in identifying discontinuities in scenarios with complex and variable rock mass structural characteristics.

Author contributions Author contributions Jiateng Guo: Conceptualization, Methodology, Resources, Supervision, Writing – original draft; Juanli Zhang: Data curation, Methodology, Software, Writing – original draft; ZiMing Xiong: Supervision, Writing – review & editing; Zirui Zhang: Methodology, Writing – review & editing; Luyuan Wang: Methodology, Writing – review & editing; Binbin Cheng: Writing – review & editing; Tianhong Yang: Supervision, Writing – review & editing.

Funding This work was financially supported by the National Natural Science Foundation of China (grant no. 42172327, 52278419, 42477197); the Fundamental Research Funds for the Central Universities (grant no. N2201022); the State Key Laboratory of Disaster Prevention & Mitigation of Explosion & Impact (grant no. LGDSKL-202209).

Data availability The data are available from the corresponding author on reasonable request.

Declarations

Competing interests The authors declare no competing interests.

References

Alejano LR, Ferrero AM, Ramírez-Oyanguren P, Fernández MÁ (2011) Comparison of limit-equilibrium, numerical and physical models of wall slope stability. Int J Rock Mech Min Sci 48:16–26. https://doi.org/10.1016/j.ijrmms.2010.06.013

Assali P, Grussenmeyer P, Villemin T, Pollet N, Viguier F (2014) Surveying and modeling of rock discontinuities by terrestrial laser scanning and photogrammetry: Semi-automatic approaches for linear outcrop inspection. J Struct Geol 66:102–114. https://doi.org/10.1016/j.jsg.2014.05.014

Aydan Ö, Ulusay R, Kawamoto T (1997) Assessment of rock mass strength for underground excavations. Int J Rock Mech Min Sci 34. https://doi.org/10.1016/S1365-1609(97)00273-6. :18.e1-18.e17

Battulwar R, Emami E, Naghadehi MZ, Sattarvand J (2020) Automatic extraction of joint orientations in rock mass using PointNet and DBSCAN. In Advances in Visual Computing: 15th International Symposium, ISVC 2020, San Diego, CA, USA, October 5–7, 2020, Proceedings, Part II 15 718–727. https://doi.org/10.1007/978-3-030-64559-5_57

Cai X, Lü Q, Zheng J, Liao KW, Liu J (2023) An efficient adaptive approach to automatically identify rock discontinuity parameters using 3D point cloud model from outcrops. Geol J 58:2195–2210. https://doi.org/10.1002/gj.4708

Cardia S, Palma B, Langella F, Pagano M, Parise M (2023) Alternative methods for semi-automatic clusterization and extraction of discontinuity sets from 3D point clouds. Earth Sci Inf 16:2895–2914. https://doi.org/10.1007/s12145-023-01029-0

Chen J, Zhu H, Li X (2016) Automatic extraction of discontinuity orientation from rock mass surface 3D point cloud. Comput Geosci 95:18–31. https://doi.org/10.1016/j.cageo.2016.06.015

Chen Q, Ge Y, Tang H (2024) An unsupervised method for rock discontinuities rapid characterization from 3D point clouds under noise. Gondwana Res 132:287–308. https://doi.org/10.1016/j.gr.2024.04.017

Ding Q, Huang R, Wang F, Chen J, Wang M, Zhang X (2018) Multi-Parameter Dominant Grouping of Discontinuities in Rock Mass Using Improved ISODATA Algorithm. Math Probl Engineerin 2018:5619404. https://doi.org/10.1155/2018/5619404

Dube D, Zell A (2011) Real-time plane extraction from depth images with the randomized hough transform. In 2011 IEEE international conference on computer vision workshops (ICCV Workshops). 2011:1084–1091. https://doi.org/10.1109/ICCVW.2011.6130371

Feng Q, An X, Ju G, Luo H, Liu J, Xu X (2021) Identification and risk assessment of a dangerous rock mass in high-steep slope based on 3D laser scanning. 2021 7th International Conference on Hydraulic and Civil Engineering & Smart Water Conservancy and Intelligent Disaster Reduction Forum (ICHCE & SWIDR) 1139–1144. https://doi.org/10.1109/ICHCESWIDR54323.2021.9656259

Ferrero AM, Forlani G, Roncella R, Voyat HI (2009) Advanced geostructural survey methods applied to rock mass characterization. Rock Mech Rock Eng 42:631–665. https://doi.org/10.1007/s00603-008-0010-4

García-Cortés S, Galán CO, Argüelles-Fraga R, Díaz AM (2012) Automatic detection of discontinuities from 3D point clouds for the stability analysis of jointed rock masses. In 2012 18th International Conference on Virtual Systems and Multimedia 595–598. https://doi.org/10.1109/VSMM.2012.6365985

Ge Y, Tang H, Xia D, Wang L, Zhao B, Teaway JW, Chen H, Zhou T (2018) Automated measurements of discontinuity geometric properties from a 3D-point cloud based on a modified region growing algorithm. Eng Geol 242:44–54. https://doi.org/10.1016/j.enggeo.2018.05.007

Ge Y, Cao B, Tang H (2022) Rock discontinuities identification from 3D point clouds using artificial neural network. Rock Mech Rock Eng 55:1705–1720. https://doi.org/10.1007/s00603-021-02748-w

Ge Y, Wang H, Liu G, Chen Q, Tang H (2025) Automated Identification of Rock Discontinuities from 3D Point Clouds Using a Convolutional Neural Network. Rock Mech Rock Eng 58:3683–3700. https://doi.org/10.1007/s00603-024-04351-1

Gigli G, Casagli N (2011) Semi-automatic extraction of rock mass structural data from high resolution LIDAR point clouds. Int J Rock Mech Min Sci 48:187–198. https://doi.org/10.1016/j.ijrmms.2010.11.009

Gigli G, Frodella W, Garfagnoli F, Morelli S, Mugnai F, Menna F, Casagli N (2014) 3-D geomechanical rock mass characterization for the evaluation of rockslide susceptibility scenarios. Landslides 11:131–140. https://doi.org/10.1007/s10346-013-0424-2

Gomes RK, de Oliveira LPL, Gonzaga L Jr, Tognoli FMW, Veronez MR, de Souza MK (2016) An algorithm for automatic detection and orientation estimation of planar structures in LiDAR-scanned outcrops. Comput Geosci 90:170–178. https://doi.org/10.1016/j.cageo.2016.02.011

Goodman RE (1995) Block theory and its application. Geotechnique 45:383–423. https://doi.org/10.1680/geot.1995.45.3.383

Guo J, Liu S, Zhang P, Wu L, Zhou W, Yu Y (2017) Towards semi-automatic rock mass discontinuity orientation and set analysis from 3D point clouds. Comput Geosci 103:164–172. https://doi.org/10.1016/j.cageo.2017.03.017

Hammah RE, Curran JH (1998) Fuzzy cluster algorithm for the automatic identification of joint sets. Int J Rock Mech Min Sci 35:889–905. https://doi.org/10.1016/S0148-9062(98)00011-4

Hu L, Xiao J, Wang Y (2020) Efficient and automatic plane detection approach for 3-D rock mass point clouds. Multimedia Tools Appl 79:839–864. https://doi.org/10.1007/s11042-019-08189-6

Hudson JA, Priest SD (1979) Discontinuities and rock mass geometry. Int J Rock Mech Min Sci Geomech Abstracts 16:339–362. https://doi.org/10.1016/0148-9062(79)90001-9

Lato M, Diederichs MS, Hutchinson DJ, Harrap R (2009) Optimization of LiDAR scanning and processing for automated structural evaluation of discontinuities in rockmasses. Int J Rock Mech Min Sci 46:194–199. https://doi.org/10.1016/j.ijrmms.2008.04.007

Leng X, Xiao J, Wang Y (2016) A multi-scale plane-detection method based on the Hough transform and region growing. Photogram Rec 31:166–192. https://doi.org/10.1111/phor.12145

Lu G, Cao B, Zhu X, Lin Z, Bai D, Tao C, Li Y (2024) Identification of rock mass discontinuity from 3D point clouds using improved fuzzy C-means and convolutional neural network. Bull Eng Geol Environ 83:159. https://doi.org/10.1007/s10064-024-03658-1

Mineo S, Caliò D, Pappalardo G (2022) UAV-based photogrammetry and infrared thermography applied to rock mass survey for geomechanical purposes. Remote Sens 14:473. https://doi.org/10.3390/rs14030473

Olariu MI, Ferguson JF, Aiken CLV, Xu X (2008) Outcrop fracture characterization using terrestrial laser scanners: Deep-water Jackfork sandstone at Big Rock Quarry, Arkansas. Geosphere 4:247–259. https://doi.org/10.1130/GES00139.1

Pantelidis L (2009) Rock slope stability assessment through rock mass classification systems. Int J Rock Mech Min Sci 46:315–325. https://doi.org/10.1016/j.ijrmms.2008.06.003

Priest SD, Hudson JA (1976) Discontinuity spacings in rock. Int J Rock Mech Min Sci Geomech Abstracts 13:135–148. https://doi.org/10.1016/0148-9062(76)90818-4. Pergamon

Puniach E, Gruszczyński W, Stoch T, Mrocheń D, Ćwiąkała P, Sopata P, Pastucha E, Matwij W (2023) Determination of the coefficient of proportionality between horizontal displacement and tilt change using UAV photogrammetry. Eng Geol 312:106939. https://doi.org/10.1016/j.enggeo.2022.106939

Qi CR, Yi L, Su H, Guibas LJ (2017) Pointnet++: Deep hierarchical feature learning on point sets in a metric space. Adv Neural Inf Process Syst 30. https://doi.org/10.48550/arXiv.1706.02413

Raguram R, Chum O, Pollefeys M, Matas J, Frahm JM (2012) USAC: A universal framework for random sample consensus. IEEE Trans Pattern Anal Mach Intell 35:2022–2038. https://doi.org/10.1109/TPAMI.2012.257

Riquelme AJ, Abellán A, Tomás R, Jaboyedoff M (2014) A new approach for semi-automatic rock mass joints recognition from 3D point clouds. Comput Geosci 68:38–52. https://doi.org/10.1016/j.cageo.2014.03.014

Riquelme AJ, Abelian A, Tomas R (2015) Discontinuity spacing analysis in rock masses using 3D point clouds. Eng Geol 195:185–195. https://doi.org/10.1016/j.enggeo.2015.06.009

Slob S, van Knapen B, Hack R, Turner K, Kemeny J (2005) Method for automated discontinuity analysis of rock slopes with three-dimensional laser scanning. Transp Res Rec 1913:187–194. https://doi.org/10.1177/0361198105191300118

Sonmez H, Ulusay R (1999) Modifications to the geological strength index (GSI) and their applicability to stability of slopes. Int J Rock Mech Min Sci 36:743–760. https://doi.org/10.1016/s0148-9062(99)00043-1

Sturzenegger M, Stead D (2009) Close-range terrestrial digital photogrammetry and terrestrial laser scanning for discontinuity characterization on rock cuts. Eng Geol 106:163–182. https://doi.org/10.1016/j.enggeo.2009.03.004

Sun W, Wang J, Yang Y, Jin F (2021) Rock mass discontinuity extraction method based on multiresolution supervoxel segmentation of point cloud. IEEE J Sel Top Appl Earth Observations Remote Sens 14:8436–8446. https://doi.org/10.1109/JSTARS.2021.3104845

Vaswani A, Shazeer N, Parmar N, Uszkoreit J, Jones L, Gomez AN, Kaiser L, Polosukhin I (2017) Attention is all you need. Advances in neural information processing systems 30. https://doi.org/10.48550/arXiv.1706.03762

Vöge M, Lato MJ, Diederichs MS (2013) Automated rockmass discontinuity mapping from 3-dimensional surface data. Eng Geol 164:155–162. https://doi.org/10.1016/j.enggeo.2013.07.008

Wang Y, Feng HY, Delorme FÉ, Engin S (2013) An adaptive normal estimation method for scanned point clouds with sharp features.

Comput Aided Des 45:1333–1348. https://doi.org/10.1016/j.cad.2013.06.003

Wang W, Zhao W, Huang L, Vimarlund V, Wang Z (2014) Applications of terrestrial laser scanning for tunnels: a review. J Traffic Transp Eng (English Edition) 1:325–337. https://doi.org/10.1016/S2095-7564(15)30279-8

Wang X, Zou L, Shen X, Ren Y, Qin Y (2017) A region-growing approach for automatic outcrop fracture extraction from a three-dimensional point cloud. Comput Geosci 99:100–106. https://doi.org/10.1016/j.cageo.2016.11.002

Wehr A, Lohr U (1999) Airborne laser scanning—an introduction and overview. ISPRS J photogrammetry remote Sens 54:68–82. https://doi.org/10.1016/S0924-2716(99)00011-8

Wu A, Zhao W, Zhang Y, Fu X (2023) A detailed study of the CHN-BQ rock mass classification method and its correlations with RMR and Q system and Hoek-Brown criterion. Int J Rock Mech Min Sci 162:105290. https://doi.org/10.1016/j.ijrmms.2022.105290

Xing Y, Kulatilake P, Sandbak LA (2018) Effect of rock mass and discontinuity mechanical properties and delayed rock supporting on tunnel stability in an underground mine. Eng Geol 238:62–75. https://doi.org/10.1016/j.enggeo.2018.03.010

Yi X, Feng W, Wang D, Yang R, Hu Y, Zhou Y (2023) An efficient method for extracting and clustering rock mass discontinuities from 3D point clouds. Acta Geotech 18:3485–3503. https://doi.org/10.1007/s11440-023-01803-w

Zhang P, Li J, Yang X, Zhu H (2018) Semi-automatic extraction of rock discontinuities from point clouds using the ISODATA clustering algorithm and deviation from mean elevation. Int J Rock Mech Min Sci 110:76–87. https://doi.org/10.1016/j.ijrmms.2018.07.009

Zhang P, Zhao Q, Tannant DD, Ji T, Zhu H (2019) 3D mapping of discontinuity traces using fusion of point cloud and image data. Bull Eng Geol Environ 78:2789–2801. https://doi.org/10.1007/s10064-018-1280-z

Zhang J, Li H, Yang X, Jiang N, Zhou JW (2021) Quantitative assessment of rockfall hazard in post-landslide high rock slope through terrestrial laser scanning. Bull Eng Geol Environ 80:7315–7331. https://doi.org/10.1007/s10064-021-02426-9

Zhou D, Wang L, An S, Wang X, An Y (2022) Integration of unmanned aerial vehicle (UAV)-based photogrammetry and InSAR for mining subsidence and parameters inversion: A case study of the Wangjiata Mine, China. Bull Eng Geol Environ 81:343. https://doi.org/10.1007/s10064-022-02845-2

Zhou J, Chen J, Li H (2024) An optimized fuzzy K-means clustering method for automated rock discontinuities extraction from point clouds. Int J Rock Mech Min Sci 173:105627. https://doi.org/10.1016/j.ijrmms.2023.105627

Zhou M, Fu H, Zhao Y, Zhou Y, Yang T, Huang W, Hu X (2025) A new method of rock discontinuity sets using modified self-organizing mapping neural network. Earth Sci Inf 18:182. https://doi.org/10.1007/s12145-024-01678-9

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.